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## # 🧪 Lab Guide: AWS RDS MySQL Setup and Ubuntu VM Integration

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### ## ✅ Prerequisites

- AWS account
- IAM user with RDS and EC2 permissions
- Ubuntu VM (EC2 or other) with internet access
- SSH access to your Ubuntu VM
- MySQL client installed on the Ubuntu VM

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### ## 🟩 Part 1: Create RDS MySQL Database (GUI)

#### ### 1. Login to AWS Console

Navigate to: [<https://console.aws.amazon.com/>](<https://console.aws.amazon.com/>)

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#### ### 2. Go to **RDS Service**

Search for **RDS** in the AWS Management Console and open it.

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#### ### 3. Click **"Create database"**

- **Engine options**: Choose `MySQL`
- **Templates**: Select `Free tier` (if eligible)
- **DB instance identifier**: `mydb-instance`
- **Master username**: `admin`
- **Master password**: Choose a strong password  
(Save this for later)

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### ### 4. DB Instance Settings

- **DB instance size**: `db.t3.micro` (Free tier eligible)
- **Storage**: General Purpose (SSD), default 20 GiB

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### ### 5. Connectivity Settings

- **VPC**: Default VPC (or choose one if you created a custom)
- **Public access**: **Yes** (to connect from your Ubuntu VM)
- **Availability zone**: (Any or leave default)
- **VPC security group**: Choose existing or create a new one

>  If creating new:

> - Name: `rds-sg`

> - Add an inbound rule:

> - Type: MySQL/Aurora

> - Port: 3306

> - Source: Custom → your Ubuntu VM IP (or `0.0.0.0/0` for open access – not recommended in production)

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### ### 6. Additional Settings

- Leave defaults or:
  - **Initial DB name**: `sampledb`
  - **Backup**: Optional
  - **Monitoring**: Optional

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### ### 7. Click **"Create Database"**

> 🎉 It may take 5–10 minutes for RDS instance creation.

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## 🟢 Part 2: Connect from Ubuntu VM

### 1. Install MySQL client (if not installed)

```
``bash
sudo apt update
sudo apt install mysql-client -y
``
```

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### 2. Get RDS Endpoint

- Go to **\*\*RDS > Databases\*\***
- Click on your instance (e.g., `mydb-instance`)
- Copy the **\*\*Endpoint\*\*** (e.g., `mydb-instance.abc123xyz.us-east-1.rds.amazonaws.com`)

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### 3. Connect to the RDS DB

```
``bash
mysql -h your-endpoint.rds.amazonaws.com -u admin -p
``
```

When prompted, enter the password you set during creation.

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## 🟢 Part 3: Create Sample Table in RDS

Once connected, create the table:

```
```sql
CREATE DATABASE IF NOT EXISTS sampled;
USE sampled;
```

```
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) NOT NULL UNIQUE,
  dob DATE
);
```
```

Insert sample data:

```
```sql
INSERT INTO users (name, email, dob) VALUES
('Alice Smith', 'alice@example.com', '1990-05-21'),
('Bob Jones', 'bob@example.com', '1985-11-03');
```
```

Verify:

```
```sql
SELECT * FROM users;
```
```

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##  Summary

| Step  | Description                         |
|-------|-------------------------------------|
| ----- | -----                               |
| 1     | Created RDS MySQL using AWS Console |
| 2     | Allowed access via Security Group   |
| 3     | Installed MySQL client on Ubuntu    |
| 4     | Connected and tested DB             |

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## ## ⚠ Notes

- **Use tighter security groups** in production (restrict IP ranges)
- Use **SSL** for encrypted database connections if needed
- Monitor **RDS billing** if using non-free-tier options

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